

Preprocessing AutoResearch

Raman ↔ Ag-SERS comparability — GAIRA V5 Stage B0

Can leakage-safe preprocessing and explicit Ag-SERS background modelling recover cross-modal biochemical correspondence?

Candidate pipelines	120
Rejected by integrity rules	66
Corpus	479 spectra · 214 Raman / 265 Ag-SERS · 51 matched analytes
Design	5 outer × 4 inner, analyte-grouped; outer test used once
Reference baseline	ASLS + Savitzky-Golay + L2
Outcome	P4 — apparent improvement is caused by overprocessing
Frozen pipeline	none

PRINCIPAL FINDING

Cross-modal retrieval CAN be pushed well past the pre-declared success thresholds (MRR +0.097, top-1 +0.108 on held-out analytes) — but only by pipelines that strip the broad shared Ag-SERS component. Those same pipelines collapse Ag-SERS replicate agreement (0.95 → 0.62) and do not improve matched-vs-mismatched band specificity, which falls. The gain is contrast geometry, not recovered shared chemistry.

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Executive summary

120 pipelines · outcome P4 · nothing frozen

Question

Stage A/B found weak Raman↔Ag-SERS correspondence, and the spectral audit showed the Ag-SERS arm is dominated by a shared colloid background with the analyte surviving as a ~5% residual. This study asks whether physically reasonable, leakage-safe preprocessing and explicit background modelling can recover cross-modal correspondence without destroying real analyte information.

What was done

120 complete candidate pipelines across seven controlled arms (baseline correction, normalization, smoothing, replicate aggregation, Ag-SERS background models, derivatives, modality-specific pipelines, and rational combinations), evaluated under 5×4 nested analyte-grouped cross-validation. The outer test analytes — both modalities — were held out entirely and consumed exactly once. Acceptance thresholds and rejection rules were frozen before the outer test was run.

Result

1. Cross-modal retrieval CAN be improved past the pre-declared bar: the best candidate reached held-out MRR 0.464 vs baseline 0.366 (+0.097, threshold +0.08) and top-1 0.284 vs 0.176 (+0.108, threshold +0.05). A retrieval-only study would have declared success.
2. Every such gain is produced by stripping the broad shared Ag-SERS component. Ag-SERS replicate cosine collapses from 0.946 to 0.620 (and to 0.579 for the best explicit background model), far below the 0.90× integrity floor.
3. Band-level specificity never improves. Matched-minus-mismatched peak correspondence is highest for the RAW baseline (+0.035) and falls for every 'improved' pipeline (+0.014, +0.020). Of 67 candidates that improved MRR, ZERO also improved peak specificity — a result independent of any threshold choice.
4. Matched-pair cosine falls for nearly every analyte; ranks improve only because mismatched similarity falls faster once the common component is removed. This is a contrast-geometry effect, not newly revealed shared chemistry.
5. Explicit background correction does NOT destroy analyte information — held-out Ag-SERS analyte 1-NN stays at 0.877–0.916 (baseline 0.896) even when 84% of Ag-SERS variance is removed, and improves to 0.916 for scaled-mean subtraction. The analyte residual is real and retained; it simply does not align with the Raman spectra.
6. Arm D's own winner was 'no background correction', and mean-subtraction made held-out cross-modal retrieval worse (MRR 0.351 vs 0.366).

Conclusion

Outcome P4 — apparent improvement is caused by overprocessing. No preprocessing pipeline is frozen. The limitation is acquisition contrast in the Ag-colloid measurement, not the preprocessing pipeline: the analyte signal is present and reproducible but does not carry analyte-specific correspondence with powder Raman at usable strength.

Central Tradeoff

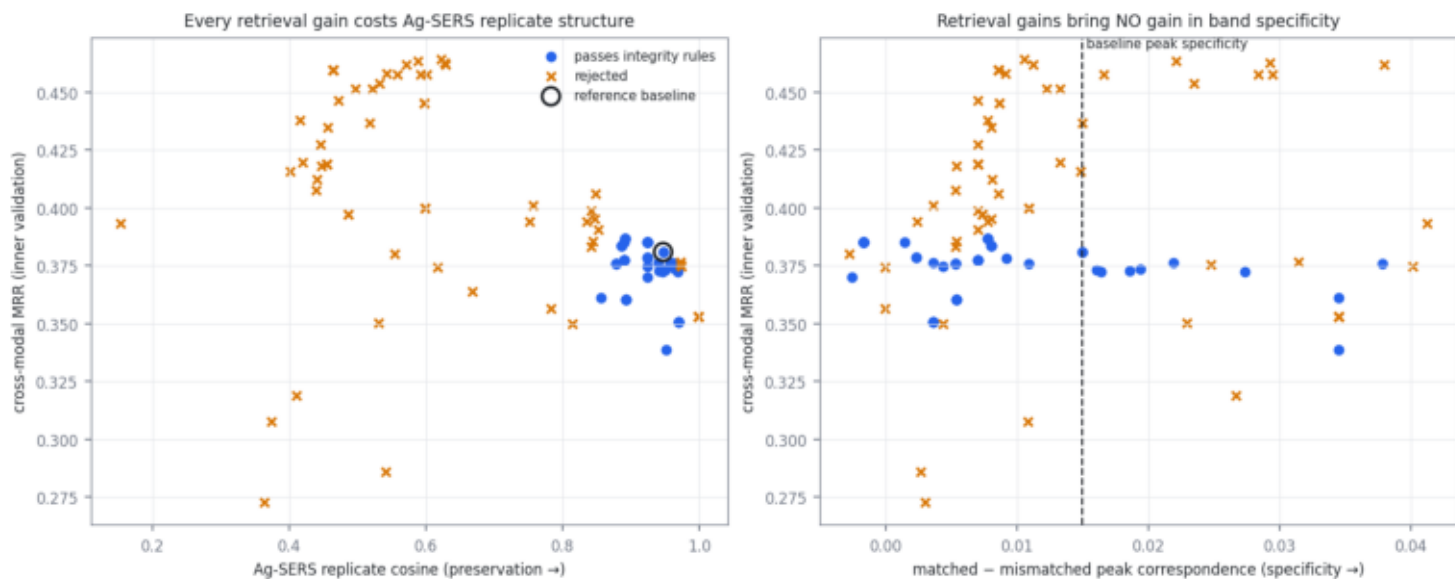


Figure 1 — The central trade-off. Left: cross-modal MRR versus Ag-SERS replicate preservation; every high-MRR candidate sits at low replicate cosine. Right: MRR versus matched-minus-mismatched peak specificity; gains in retrieval are not accompanied by gains in band specificity.

Pareto And Rejections

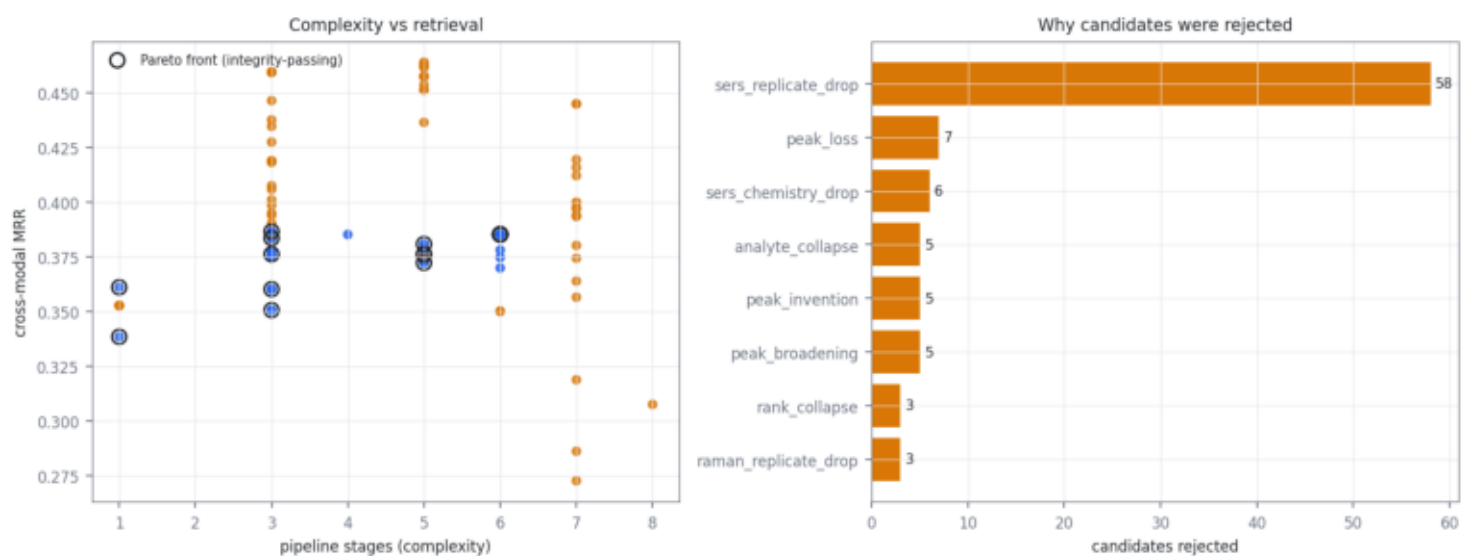


Figure 2 — Complexity versus retrieval, and the reasons candidates were rejected. Ag-SERS replicate destruction dominates (58 of 66 rejections).

Background Control

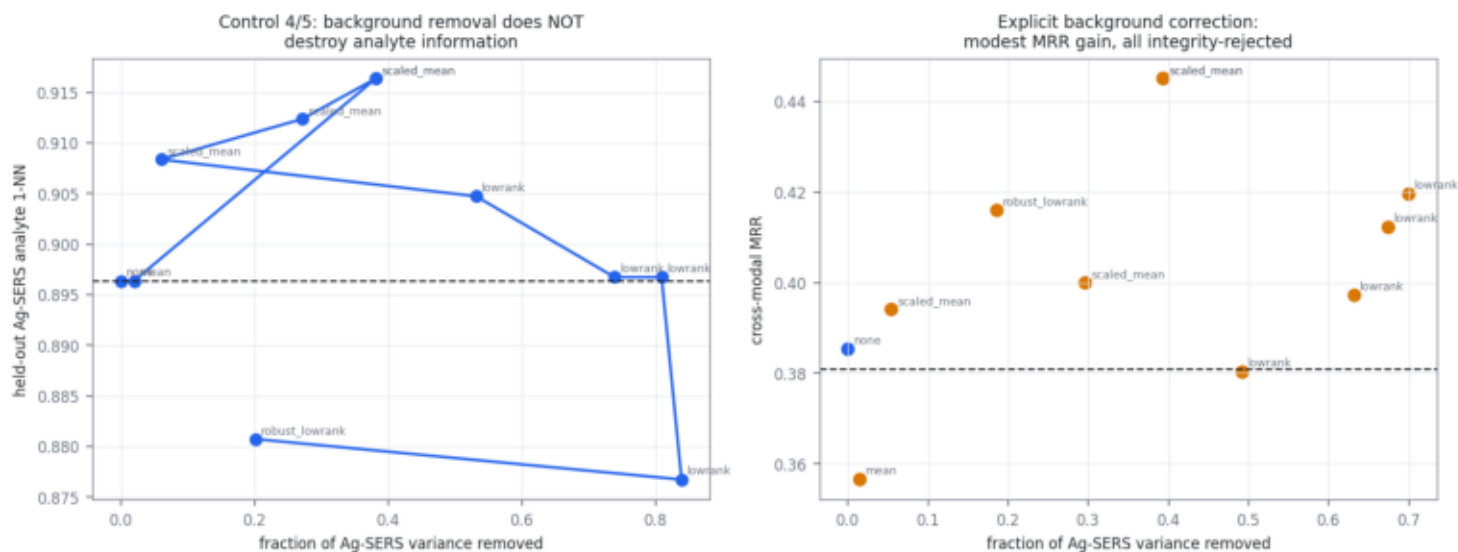


Figure 3 — Ag-SERS background controls. Left (Control 4/5): removing up to 84% of Ag-SERS variance leaves held-out analyte identification intact— the analyte residual is genuinely retained. Right: explicit background correction yields only a modest MRR gain and every variant fails the integrity rules.

Outer Test

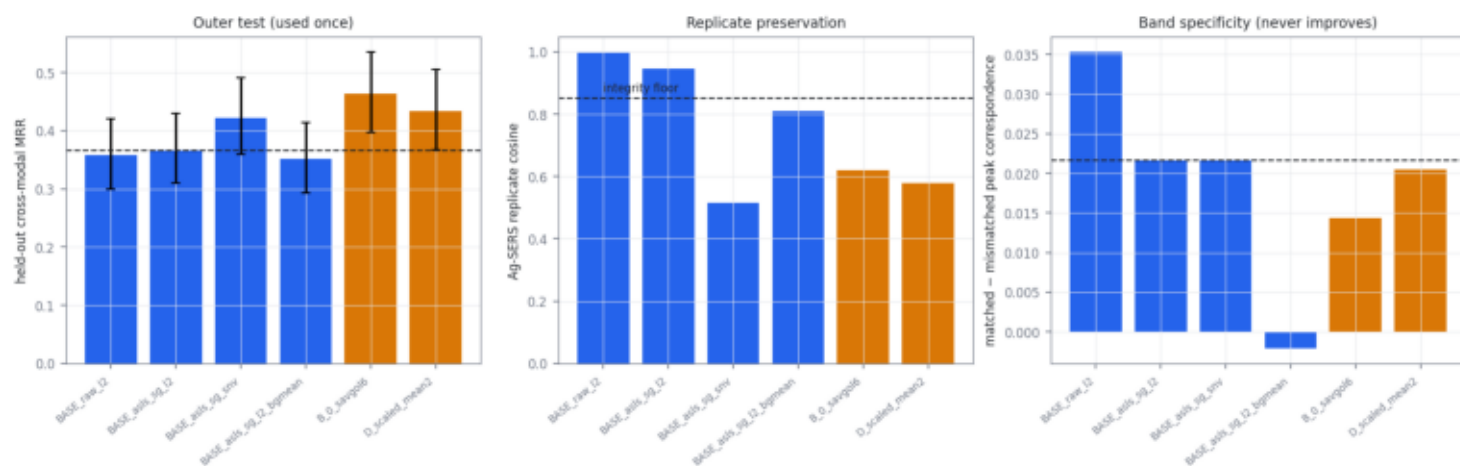


Figure 4 — The one-time outer test. Retrieval (left) rises exactly where replicate preservation (middle) collapses, while band specificity (right) never improves.

Smoothing Normalization

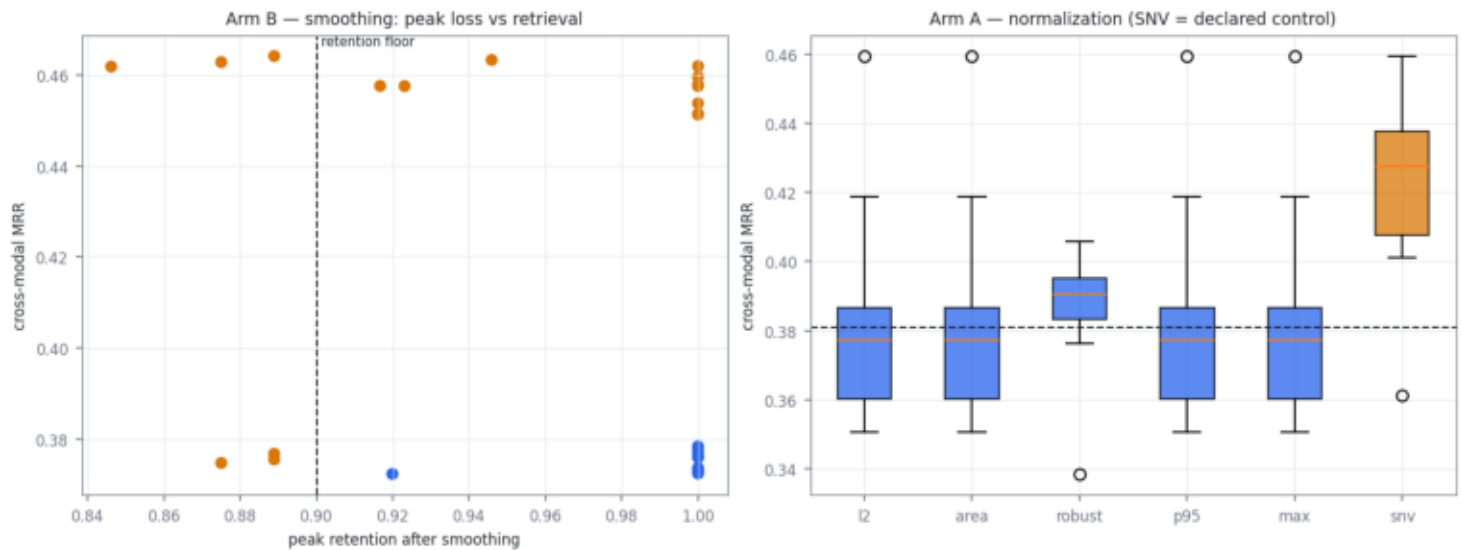


Figure 5 — Arm B (smoothing) and Arm A (normalization). Retrieval rises as peak retention falls. SNV, included only as a declared negative control, scores well on retrieval— precisely the failure mode this study was designed to catch.

Per Analyte Family

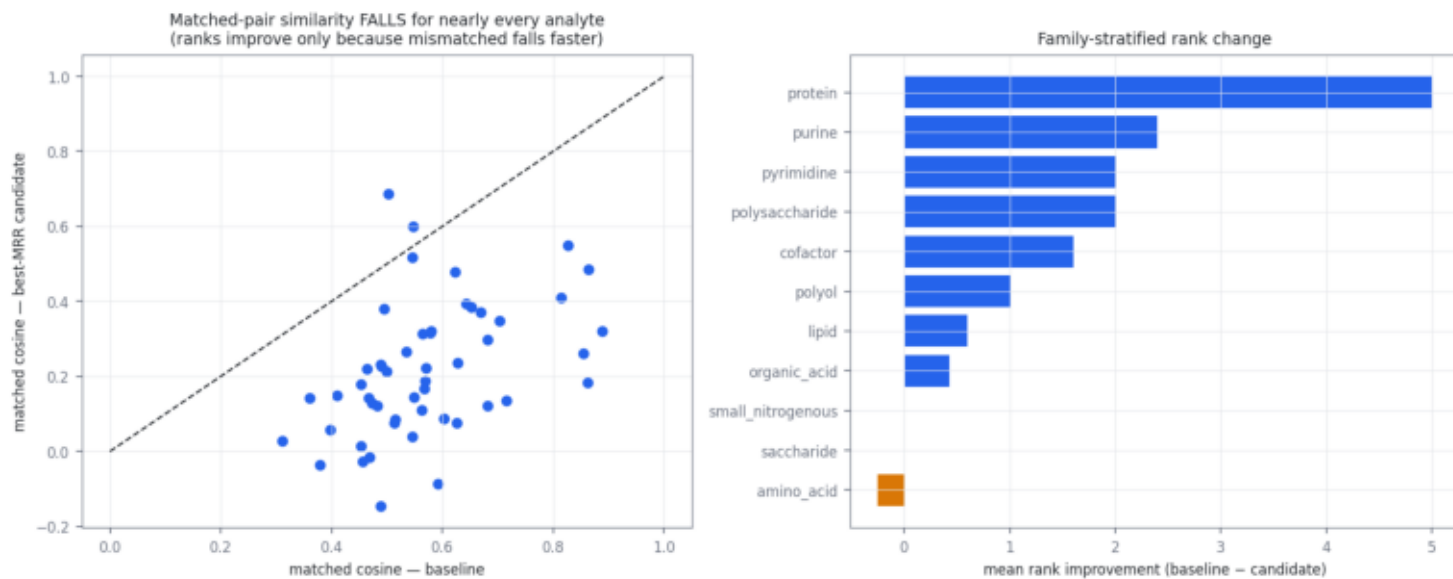


Figure 6 — Per-analyte and family results. Matched-pair cosine falls for nearly every analyte under the best-MRR candidate; rank 'improvements' come from mismatched similarity falling faster.

Fold Stability

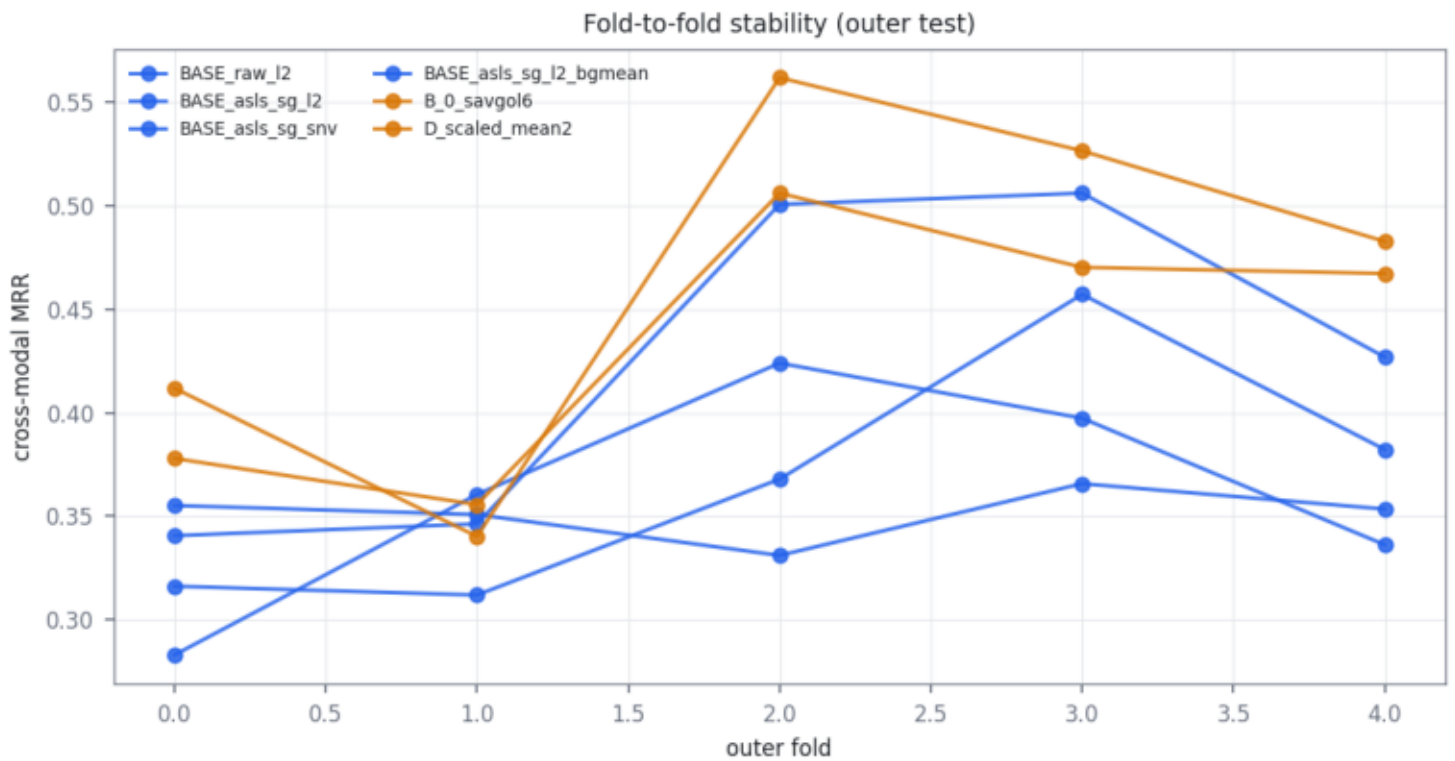


Figure 7 — Fold-to-fold stability of held-out cross-modal MRR.

Decision

Outcome P4 — apparent improvement is caused by overprocessing

Inner selection produced NO eligible candidate: of 120 pipelines, none improved cross-modal retrieval AND matched-peak specificity while satisfying the spectral-integrity rejection rules.

characterisation B_0_savgol6: $\Delta\text{MRR} +0.097$, Ag-SERS replicate cosine 0.620 vs baseline 0.946 (integrity_ok=False), $\Delta\text{peak-specificity} -0.0072$.

characterisation D_scaled_mean2: $\Delta\text{MRR} +0.068$, Ag-SERS replicate cosine 0.579 vs baseline 0.946 (integrity_ok=False), $\Delta\text{peak-specificity} -0.0011$.

Cross-modal retrieval rises only for pipelines that strip the broad shared component, which collapses Ag-SERS replicate agreement without any gain in matched-vs-mismatched peak specificity — appearance, not shared chemistry.

Pre-declared acceptance thresholds (frozen before the outer test)

outer_mrr_improvement_abs	0.08
outer_top1_improvement_abs	0.05
mrr_bootstrap_ci_excludes_zero	True
peak_effect_relative_improvement	0.5
sers_replicate_min_frac_of_L2	0.9
within_modality_retrieval_min_frac	0.9
peak_retention_min	0.9
peak_invention_max	0.02
duplicate_collapse_increase_allowed	0.0
min_fraction_outer_folds_stable	0.8
reference_baseline	BASE_asls_sg_l2

Next authorized action

No preprocessing pipeline is frozen and no representation work is authorized by this study. The evidence points to acquisition contrast rather than preprocessing: the recommended next step is targeted Ag-SERS re-acquisition in which the analyte, not the colloid, dominates the spectrum (higher effective surface coverage, blank-colloid difference measurement, or Au-SERS references), followed by re-running this same frozen study design on the enlarged corpus.

Scope

Read-only study. Stage A/B results, historical V1–V3.1 pipelines, the demo, and all governing documents were left unmodified. The outer test folds were consumed exactly once, recorded in `configs/study_manifest.json`.